Fiscal Year:	FY 2013	**Task Last Updated:**	FY 03/20/2013
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Project Title:	Novel Double-Hit Mouse Model to Investigate Oxidative Damage from Radiation/ Hyperoxia Related to Space Exploration - Evaluation of Dietary Flaxseed as Countermeasure		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-4318	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Crew Health NNJ11ZSA002NA
Start Date:	06/01/2012	**End Date:**	05/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 5/31/2014 (from 5/31/2013) per A. Chu/ARC (Ed., 3/15/2013)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	NNX12AK19G		
Performance Goal No.:			
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Task Description:	Spaceflight missions may require crewmembers to conduct extravehicular activities (EVA) for repair, maintenance or scientific purposes. EVAs should total no more than 24 hours weekly per crew member. During each EVA, crewmembers are subjected to a number of challenges that may induce oxidative tissue damage and thus present a threat to their health and therefore pose a limitation to the success of the mission. One such challenge is the exposure to 100% oxygen as well as to low levels of total body cosmic/galactic radiation that crewmembers will be exposed to during an EVA. Currently, no studies have addressed this “double hit” hyperoxia/radiation effect so we developed and characterized a novel, mouse model of repeated total body radiation and hyperoxia exposure. Preliminary studies identified acute lung damage associated with such treatment so the goal of the proposed study is to further characterize the acute damage and most importantly, to provide information on the long-term effects of such an exposure (Aim 1). Additionally, initial findings suggested that dietary administration of whole grain dietary flaxseed (FS) ameliorated adverse acute effects in the same model. Our group was the first to show in murine models of radiation and hyperoxic lung injury the protective properties of FS. Importantly, lignan phenolics in FS such as secoisolariciresinol diglucoside (SDG), are known antioxidant, anti-inflammatory and anticarcinogenic agents. We hypothesize that FS and FS lignans will be effective countermeasures of this double-hit toxicity to lung tissues (Aim 2). We plan to further explore the usefulness of FS and SDG in both acute and chronic oxidative lung damage in this model. FS and specifically SDG are safe, readily available and thus, attractive candidates to evaluate as oxidative damage countermeasures in the proposed double hit radiation/hyperoxia model. Data from this work are relevant to EVA-related pathophysiology associated with space exploration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We aim to identify novel risks to lung tissues as a result of exposure to hyperoxia and radiation. This is relevant to both space travel and clinical Earth scenarios. We hope that our proposed countermeasure will mitigate risk.
Task Progress:	RESEARCH GRANT IDENTIFIER: NNX12AK19G DATE: 3/15/2013 ANNUAL PROGRESS REPORT: Since the start of the award period, we have completed Specific Aim 1 and have initiated Aim 2 (currently on-going). Our results were presented at NASA’s annual Human Research Program workshop at Galveston, Texas earlier this year. Results have been described in a manuscript to be submitted to Radiation Research by the end of April. We have requested a no-cost extension to complete Aim 2 and prepare relevant manuscripts. Specific Aim 1: Characterize oxidative lung damage from repeated Radiation/hyperoxia exposure in a novel mouse model of space-flight related challenge. Rationale: Spaceflight missions may require crewmembers to conduct extravehicular activities (EVA) for repair, maintenance or scientific purposes. Pre-breathe protocols in preparation for an EVA entail 100% hyperoxia exposure that may last for a few hours (5-8 hours), which may be repeated 2-3 times weekly. Each EVA is associated with additional challenges such as low levels of total body cosmic/galactic radiation exposure that may present a threat to crewmember health and therefore, pose a threat to the success of the mission. We have developed a murine model of combined, acute hyperoxia and radiation exposure (double-hit) in the context of evaluating countermeasures to oxidative lung damage associated with space flight. In the current study, our objective was to characterize the long-term effects of a double-hit challenge on lung tissue using a novel murine model of repeated exposure to low-level radiation and hyperoxia. This is the first study of its kind evaluating chronic lung damage relevant to space exploration in a rodent model. Conclusions/Significance of Findings: We have characterized significant, chronic lung changes in our murine model of repeated radiation and hyperoxia exposure relevant to space travel. Lung tissue changes, detectable several months after the original exposure, include significant oxidative lung damage (lipid peroxidation) and increased pulmonary fibrosis. These findings, along with the observed decreases in blood oxygenation levels in all challenge conditions (whether single or in combination), lead us to conclude that in our model of repeated exposure to oxidative stressors, chronic tissue changes are detected that persist even months after the exposure to the stressor has ended. This data will provide useful information in the design of countermeasures to tissue oxidative damage associated with space exploration. Specific Aim 2: Evaluate usefulness of dietary FS and FLC as countermeasure to oxidative lung injury in a mouse model of radiation/hyperoxia relevant to space flight challenges. Rationale: Preliminary data from our team showed that whole grain dietary flaxseed (FS) boosts antioxidant defense in murine tissues by enhancing antioxidant enzyme expression. Importantly, dietary FS ameliorates adverse effects of acute tissue injury resulting from hyperoxia and ischemia/reperfusion oxidative damage as well as from thoracic radiation by enhancing survival, decreasing inflammation and blocking long term tissue fibrosis1-5. Flaxseed contains lignans such as the phenolic secoisolariciresinol diglucoside (SDG), compounds with known antioxidant and anti-inflammatory properties. We hypothesize that flaxseed, via the action of its phenolic lignan component; will be an effective countermeasure to radiation/hyperoxia-induced acute and chronic oxidative tissue damage. Data from this work are relevant to space flight-related pathophysiology associated with space exploration. Approach: Mouse cohorts were placed on a flaxseed (FS) or flaxseed lignan component (FLC) diets. Lung tissues, as well as brain, and blood will be collected after 1 month (acute) and compared to that collected after 2, 3, 4 months (Chronic) with respect to oxidative damage, inflammation, fibrosis. Results: Pending.
Bibliography Type:	Description: (Last Updated: 10/31/2019)
Abstracts for Journals and Proceedings	Pietrofesa RA, Arguiri E, Solomides CC, Christofidou-Solomidou M. "Oxidative lung damage resulting from prolonged exposure to radiation and hyperoxia associated with space exploration." 2013 NASA Human Research Program Investigators’ Workshop, Galveston, TX, February 12-14, 2013. 2013 NASA Human Research Program Investigators’ Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013